

Regularisation and early stopping

Project: Image-pattern Network Investigation

1. Prepare the purpose, prerequisites and materials

Teach the learning cycle

Primary target: Separate a weight penalty from prediction error

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Training and validation curves

Without opening the old lesson, explain one key idea from Training and validation curves. Give an example and a mistake that the idea helps you avoid.

Look for: Compute comparable training and validation losses at saved checkpoints

Reference example, not the only acceptable wording: Teacher explanation: Predictions [3, 4], residuals [-1.5, -2], squares [2.25, 4], MSE 3.125.

Batching and epochs

Without opening the old lesson, explain one key idea from Batching and epochs. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish an example a batch an update and an epoch

Reference example, not the only acceptable wording: Teacher explanation: Batch lengths 4, 4, 2; three updates per epoch, nine total updates and 30 example visits.

Gradients with a numerical check

Without opening the old lesson, explain one key idea from Gradients with a numerical check. Give an example and a mistake that the idea helps you avoid.

Look for: Estimate one partial derivative while other parameters stay fixed

Reference example, not the only acceptable wording: Teacher explanation: Prediction 3, residual 0, loss 0, gradient [0, 0].

Use two 45-minute sessions. Keep prediction error, penalty and total objective in separate columns. Teach early stopping as a second mechanism with a precise state machine on paper. Cover future curve values until a decision is made; do not reward retrospective selection as if it were a live stopping rule.

Outcomes

Separate a weight penalty from prediction error

Trace a regularised update under a stated objective

Apply a stopping rule and restore the best observed checkpoint

Preparation

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Prerequisite answer

Restore the parameters saved at the chosen checkpoint. The last parameters produced by training may have a different validation loss.

End goal for the pictorial case

By the end, I can explain regularisation and early stopping using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain and model the mechanism

Keep $h = \text{ReLU}(x)$, $y = vh$, train only v on $(1, 2)$ and $(2, 4)$, and freeze all hidden parameters and the zero output bias. Training MSE is $L(v) = 2.5(v - 2)^2$. Define $J(v) = L(v) + \lambda v^2$, with λ nonnegative. Only this one output weight is penalised here.

For $\lambda = 0.5$, the penalty at $v = 1.5$ is $0.5 \times 2.25 = 1.125$. Prediction MSE is 0.625 , so total objective J is 1.75 . Keep these quantities separate in logs. A smaller total objective does not automatically mean a smaller prediction error.

Among these three candidates, the smallest penalised objective is at $v = 1.5$, whereas the smallest training MSE is at $v = 2$. Penalising a weight changes the tradeoff. It does not guarantee better validation performance or fair outcomes.

The data gradient is $5(v - 2)$. The penalty gradient is $2 \times \lambda v$, giving total gradient $5(v - 2) + 2 \lambda v$. At $v = 1.5$ and $\lambda = 0.5$, these contributions are -2.5 and 1.5 , so total gradient is -1 . A rate 0.1 step gives $v = 1.6$.

Assume finite numeric values and nonnegative penalty. At the new v , data loss is 0.4 , penalty 1.28 and total 1.68 . Choose penalty strength using a declared validation prediction metric, not by comparing total objectives with different λ s as if they were the same measure.

Term · Meaning

Term: Regularisation

Meaning: A constraint or training procedure intended to limit fitting choices and improve generalisation

Term: Weight penalty

Meaning: An added objective term that discourages specified large parameter values

Term: Patience

Meaning: The allowed run of consecutive non-improving evaluations before stopping under a stated rule

Term: Restore

Meaning: Reload the parameters saved at the selected checkpoint

Patience two with strict improvement

Epoch · Validation MSE · Decision

Epoch: 0 then 1 then 2

Validation MSE: 4 then 2 then 1.5

Decision: Save each improved state

Epoch: 3 then 4

Validation MSE: 1.6 then 1.4

Decision: Wait once then reset

Epoch: 5 then 6

Validation MSE: 1.45 then 1.46

Decision: Two misses so stop at 6

Initial checkpoint is epoch 0. Evaluate once after each epoch. A strict decrease is improvement; an exact tie is a miss. Reset the miss count after improvement and stop at two consecutive misses. Restore epoch 4, the best observed checkpoint.

Guided: miss counts at epochs 3, 4, 5, 6 are 1, 0, 1, 2. Stop at 6 and restore 4. Exit: using final parameters ignores restoration; selecting patience with test losses uses test data for development. Restore the selected checkpoint, choose policy using validation, then freeze it before evaluating suitable untouched final examples.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Patience two with strict improvement

Epoch · Validation MSE · Decision

Epoch: 0 then 1 then 2

Validation MSE: 4 then 2 then 1.5

Decision: Save each improved state

Epoch: 3 then 4

Validation MSE: 1.6 then 1.4

Decision: Wait once then reset

Epoch: 5 then 6

Validation MSE: 1.45 then 1.46

Decision: Two misses so stop at 6

Initial checkpoint is epoch 0. Evaluate once after each epoch. A strict decrease is improvement; an exact tie is a miss. Reset the miss count after improvement and stop at two consecutive misses. Restore epoch 4, the best observed checkpoint.

At epochs 3, 4, 5 and 6, record the miss count after evaluation. Which epoch stops training, and which epoch supplies the restored parameters?

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Regularisation and early stopping

Separate a regularisation penalty from prediction loss and an early-stopping rule.

The facts we will use

Toy objective $J(a)=(a-3)^2+2a^2$; start $a = 0$; gradient $6a-6$; rate $1/6$. Separate validation losses:[5, 3, 3, 4], patience 1, no ties as improvement.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Regularisation and early stopping

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Regularised update

At 0 gradient -6; update gives $a = 1$.



2. Split objective

At 1 prediction term 4, penalty 2, total 6; at 0 total 9.



3. Early-stop separately

Best validation checkpoint is 2 with loss 3; checkpoint 3 is no improvement, so stop at 3.

CHANGE ONE THING

Change patience from 1 to 2 under the same tie rule.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Regularised update

What happens and why: At 0 gradient -6; update gives $a = 1$.

Step: 2. Split objective

What happens and why: At 1 prediction term 4, penalty 2, total 6; at 0 total 9.

Step: 3. Early-stop separately

What happens and why: Best validation checkpoint is 2 with loss 3; checkpoint 3 is no improvement, so stop at 3.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Regularisation adds a parameter penalty to the training objective. Here it discourages large a , so the objective balances prediction error against that penalty. Early stopping is a different decision based on the supplied validation sequence. A tied validation loss is not improvement under this policy; stop when the patience allowance is used and restore the best checkpoint.

What this example does not establish

Regularisation and early stopping are different tools; neither guarantees better performance in every dataset.

Change one thing in the pictured case

Change: Change patience from 1 to 2 under the same tie rule.

Prediction: Stop at checkpoint 4; still restore checkpoint 2.

Reason: Checkpoints 3 and 4 are two consecutive non-improvements; a later stop does not redefine the best checkpoint.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Time or stage · Teacher action

Time or stage: Session 1 start to 7 minutes

Teacher action: Retrieve checkpoint restoration and introduce the named penalised weight.

Time or stage: 7 to 18 minutes

Teacher action: Calculate the objective table and $W1$, identifying the nonzero penalty gradient at $v = 2$.

Time or stage: 18 to 32 minutes

Teacher action: Trace the worked regularised update and use W2 for the next step.

Time or stage: 32 to 45 minutes

Teacher action: Run or trace the code and distinguish objective reduction from validation evidence.

Time or stage: Session 2 start to 8 minutes

Teacher action: Reveal the validation sequence one epoch at a time; complete the guided miss counts.

Time or stage: 8 to 20 minutes

Teacher action: Complete W3 before revealing the future value; compare stopping and restored epochs.

Time or stage: 20 to 35 minutes

Teacher action: Collect W4 independently, scoring the arithmetic and the tied-loss policy separately.

Time or stage: 35 to 45 minutes

Teacher action: Discuss W5 and the exit response, then use the fresh retry and record the complete policy.

Use the worked example in Sections 3 and 6. Collect a prediction before revealing its result; ask the learner to point to each intermediate fact.

Use a ledger with columns epoch, metric, best epoch and miss count; cover future rows. Extension: solve $5(v - 2) + 2\lambda v = 0$ to obtain $v = 10 / (5 + 2\lambda)$. For nonnegative λ , discuss why the optimum moves toward zero without claiming this always improves prediction.

Reduce help deliberately

Completion task: Start $v = 1.6$ with $\lambda = 0.5$ and rate 0.1. Calculate both gradient contributions, new v and new prediction MSE.

Teacher check: Teacher explanation: Data gradient -2, penalty gradient 1.6, total -0.4. New $v = 1.64$; prediction MSE is 2.5 times 0.36 squared = 0.324.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is

understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Regularisation and early stopping. Proposed explanation: “Zero prediction error means no regularised update”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: The penalty gradient can remain nonzero.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Why should the penalty value not be reported as validation error? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Reasoned teacher answer

It is part of the training objective, not the measured validation prediction loss. Report these separate quantities with their definitions. Changed-case check: Stop at checkpoint 4; still restore checkpoint 2. Checkpoints 3 and 4 are two consecutive non-improvements; a later stop does not redefine the best checkpoint.

Changed-case reasoning: Stop at checkpoint 4; still restore checkpoint 2. Checkpoints 3 and 4 are two consecutive non-improvements; a later stop does not redefine the best checkpoint.

Collect each learner’s explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the investigation and diagnose misconceptions

Zero prediction error means no regularised update: The penalty gradient can remain nonzero.

Early stopping adds a weight penalty: It limits training duration using a validation rule; a weight penalty changes the objective.

Restore the epoch where stopping happened: Restore the best observed checkpoint under the declared rule, which can be earlier.

Use Section 9 for the evidence ledger. Add a boundary or missing-input case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Guided and exit questions

Guided: miss counts at epochs 3, 4, 5, 6 are 1, 0, 1, 2. Stop at 6 and restore 4. Exit: using final parameters ignores restoration; selecting patience with test losses uses test data for development. Restore the selected checkpoint, choose policy using validation, then freeze it before evaluating suitable untouched final examples.

W1 Separate the objective terms

At $v = 2$ and $\lambda = 0.5$ in the lesson, compute prediction MSE, penalty, total objective and total gradient. Does zero training MSE make this gradient zero?

Teacher explanation: MSE 0, penalty 2, total J 2, total gradient 2. Zero data gradient does not remove the penalty gradient.

W2 Trace the next update

Start $v = 1.6$ with $\lambda = 0.5$ and rate 0.1. Calculate both gradient contributions, new v and new prediction MSE.

Teacher explanation: Data gradient -2, penalty gradient 1.6, total -0.4. New $v = 1.64$; prediction MSE is 2.5 times 0.36 squared = 0.324.

W3 Apply patience without peeking

Validation MSE at epochs 0 to 7 is [4, 2, 1.5, 1.6, 1.4, 1.45, 1.46, 1.3]. Apply the lesson policy with patience 2. Give the stop epoch, restored epoch and why the listed epoch 7 cannot change this decision.

Teacher explanation: Stop at epoch 6 after misses at 5 and 6; restore epoch 4 with MSE 1.4. A live run has stopped before observing epoch 7; hindsight cannot be included in that policy decision.

W4 Transfer both rules independently

For a new objective $J(a) = (a - 2)^2 + a^2$, gradient $4a - 4$, start $a = 0$ and take one step at rate 0.25. Give new a , prediction error term, penalty and total J . Separately apply patience 2 with strict improvement to validation values [3, 2, 2, 2, 1] at epochs 0 to 4; report the stop and restored epochs.

Teacher explanation: The update gives $a = 1$. Prediction error 1, penalty 1, total J 2. For the separate curve, epoch 1 is best at 2; ties at 2 and 3 are two misses, so stop at epoch 3 and restore epoch 1. Epoch 4 is not observed by that live policy.

W5 Compare penalties fairly

A team compares $\lambda = 0$ and $\lambda = 0.5$ by choosing whichever has the smaller penalised training objective. Explain why this does not establish the better prediction model and name a suitable development comparison.

Teacher explanation: The objectives use different penalties, and both are training quantities. Compare the candidates on the same fixed validation examples using the same declared prediction metric, such as ordinary MSE. Reserve final test examples from setting selection.

Independent assessment and fresh retry

W4 requires $a = 1$ and terms 1, 1, 2 plus stop epoch 3 and restored epoch 1. Fresh retry: $J(b) = (b - 3)^2 + 0.5b^2$, gradient $3b - 6$; start $b = 0$, rate 0.25. New $b = 1.5$, data term 2.25, penalty 1.125, total 3.375. On validation [5, 3, 3.2, 3.1] with patience 2, stop at 3 and restore 1.

Assess independent transfer

Selected case: For a new objective $J(a) = (a - 2)^2 + a^2$, gradient $4a - 4$, start $a = 0$ and take one step at rate 0.25. Give new a , prediction error term, penalty and total J . Separately apply patience 2

with strict improvement to validation values [3, 2, 2, 2, 1] at epochs 0 to 4; report the stop and restored epochs.

Selected case answer

Teacher explanation: The update gives $a = 1$. Prediction error 1, penalty 1, total J 2. For the separate curve, epoch 1 is best at 2; ties at 2 and 3 are two misses, so stop at epoch 3 and restore epoch 1. Epoch 4 is not observed by that live policy.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect the project, support and next lesson

Add a labelled regularisation and early stopping evidence sheet to Image-pattern Network Investigation. Show how you can separate a weight penalty from prediction error. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Use a ledger with columns epoch, metric, best epoch and miss count; cover future rows. Extension: solve

$5(v - 2) + 2\lambda v = 0$ to obtain $v = 10 / (5 + 2\lambda)$. For nonnegative λ , discuss why the optimum moves toward zero without claiming this always improves prediction.

Week 17 investigates initialisation and reproducibility.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: ai-5/17, ai-5/19, ai-5/22. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.